

IMCA 221
Programming for Artists
Winter 2025

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**Class is on Zoom even in the classroom
for sharing, find the details on Moodle**

<https://moodle.concordia.ca/>

Download the slides!

Final projects - April 10

Prototype - March 27

- If you don't pay your fees you can't reserve rooms to present, **PAY YOUR FEES!!!!**

- Decide if you are going to work alone or in teams, this will determine our critique length

- Sign up here (times are not final)

<https://docs.google.com/spreadsheets/d/1tYaLwQvdHKLPFb9jIBjtOkV2yulljyn--IB8jXvJ52s/edit?gid=0#gid=0>

Final projects - April 10

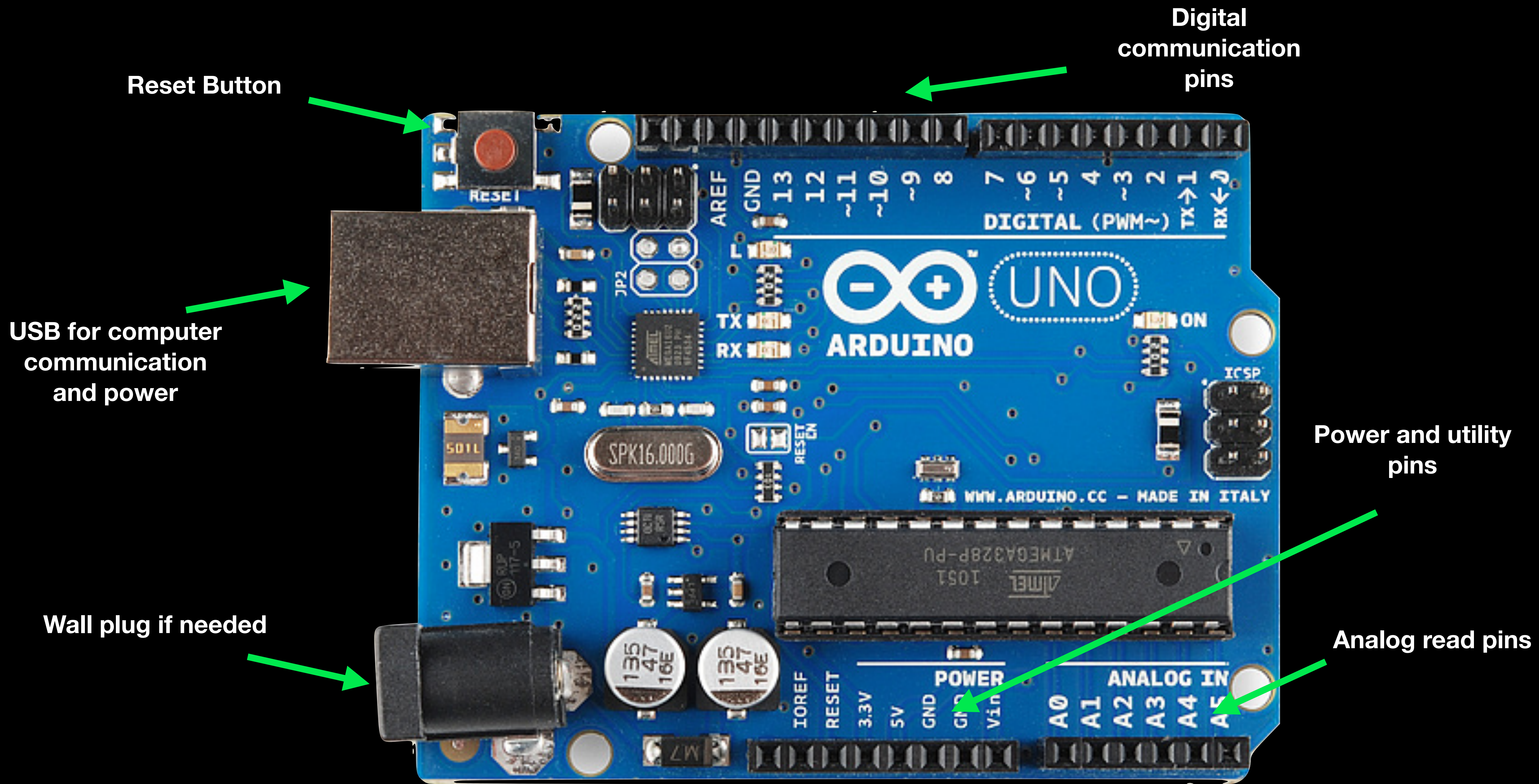
Prototype - March 27

Prototype should include:

- A working idea to show the class
- A written timeline of work to be completed (What will happen, when
- List of equipment needed for your final (things to make, cords, adapters, projectors, mounting supplies etc)
- 2 or 3 questions to ask the class during critique session

Arduino
Micro controller
platform for
controlling
physical things





Serial

We will be using a serial communication protocol to communicate over USB to the patch. You can send any message over Serial and decode it at the end.

There are lots of ways to do this, you can use wifi, midi, bluetooth or other techniques.

MQTT <https://docs.arduino.cc/tutorials/uno-wifi-rev2/uno-wifi-r2-mqtt-device-to-device/>

Arduino MIDI over USB <https://www.instructables.com/DIY-USB-Midi-Controller-With-Arduino-a-Beginners-G/>

Analog Sensors

- Provide a range of numbers according to an input.
- Light sensors (LDR)
- Pressure sensors (FSR)
- Flex Sensors
- Knobs



Buttons (normal)

- Push buttons
- Switches
- Tilt Switch
(changes as you turn it)



Buttons (weird)

- You can make buttons out of anything conductive

Great resource:

<https://>

www.kobakant.at/

[DIY/](https://www.kobakant.at/DIY/)



Range Finders & Accelerometers

- Range finders indicates a distance between the sensor and nearest object in line of sight
- accelerometer can be used to determine approximate orientation and speed



So many other sensors!

- Heart rate, GPS,
temperature, humidity,
sound, vibration,
atmospheric, gyroscope...
check out more at:

- [https://www.adafruit.com/
category/35](https://www.adafruit.com/category/35) &

- [https://www.sparkfun.com/
categories/23](https://www.sparkfun.com/categories/23)

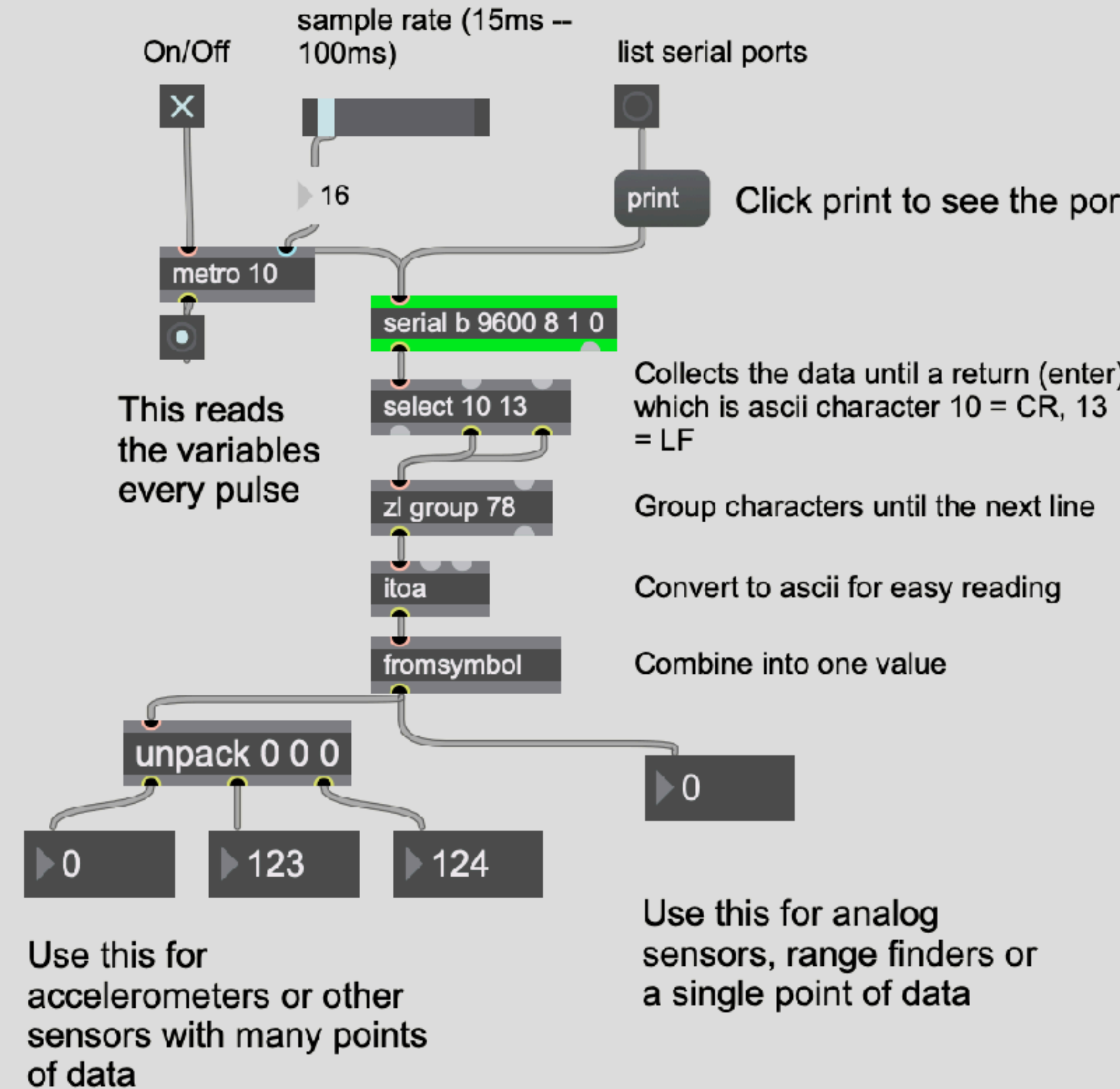


Using Sensors

- I've pre-loaded Arduino with code corresponding to sensors, they are all labeled.
- Plug in each sensor according to the instructions
- Open the Arduino-to-max patch

Arduino-To-Max patch

- Plug in the Arduino
- Press Print to see ports
- Find the port letter that says Arduino
- Put that letter in the green box (before 9600)
- Connect your circuit
- Toggle on/off

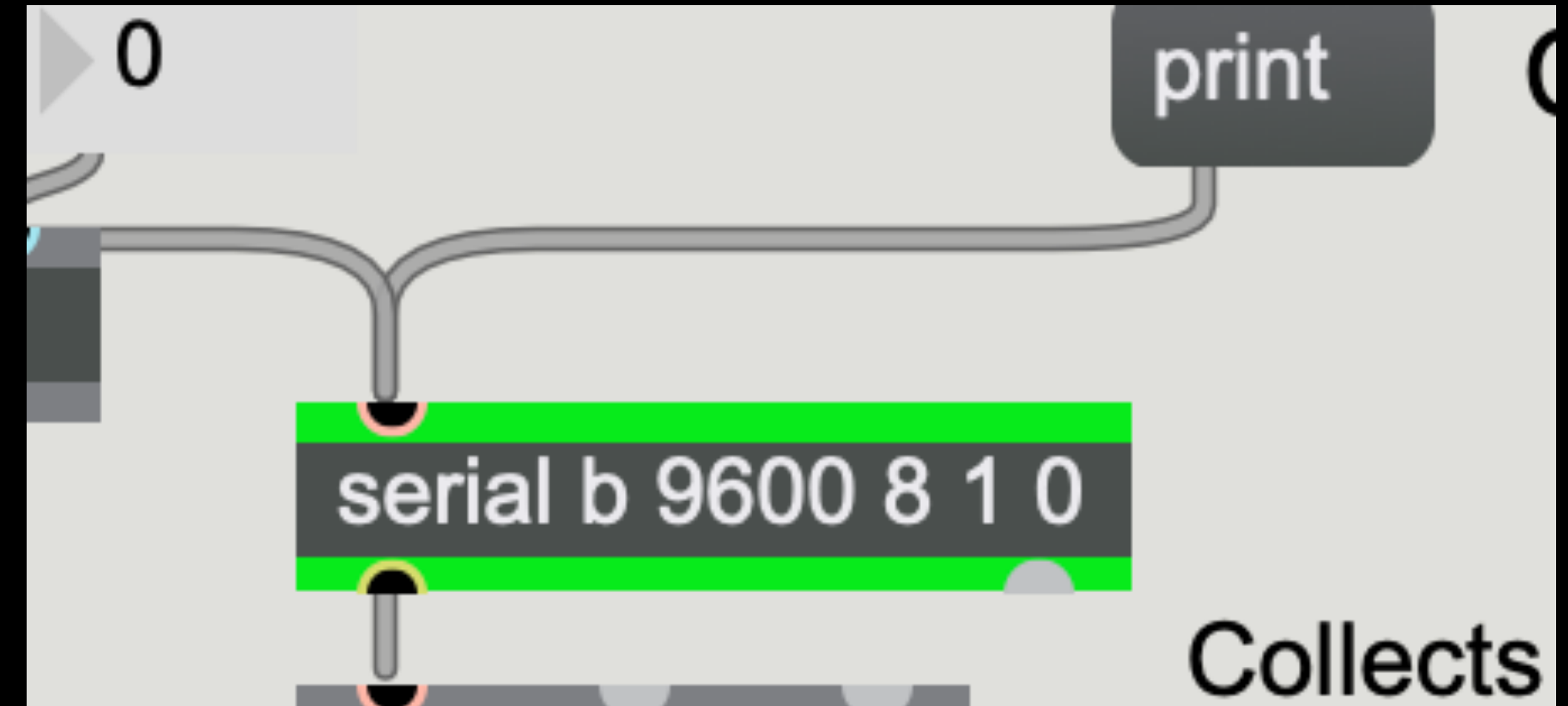
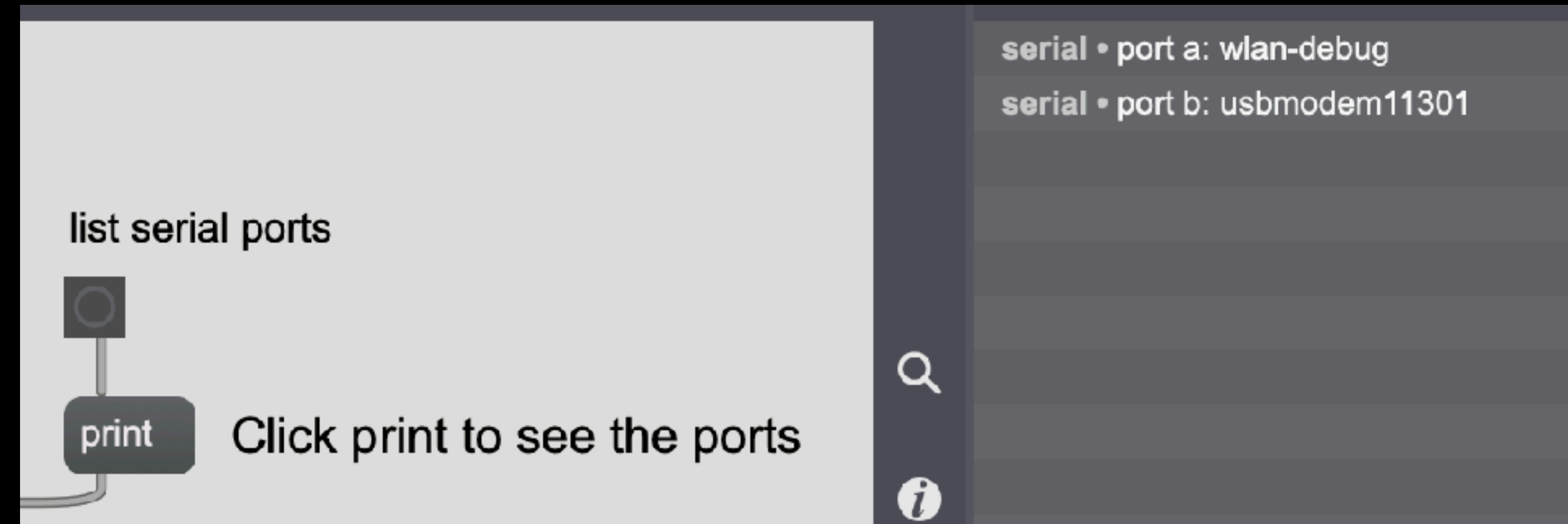


Arduino-To-Max patch

- Print will show all ports available in the **print log**.

11301 or **Arduino Uno** is usually the one we want, depending on the model you have.

- Add the port letter in the green box. Here it is **B**.



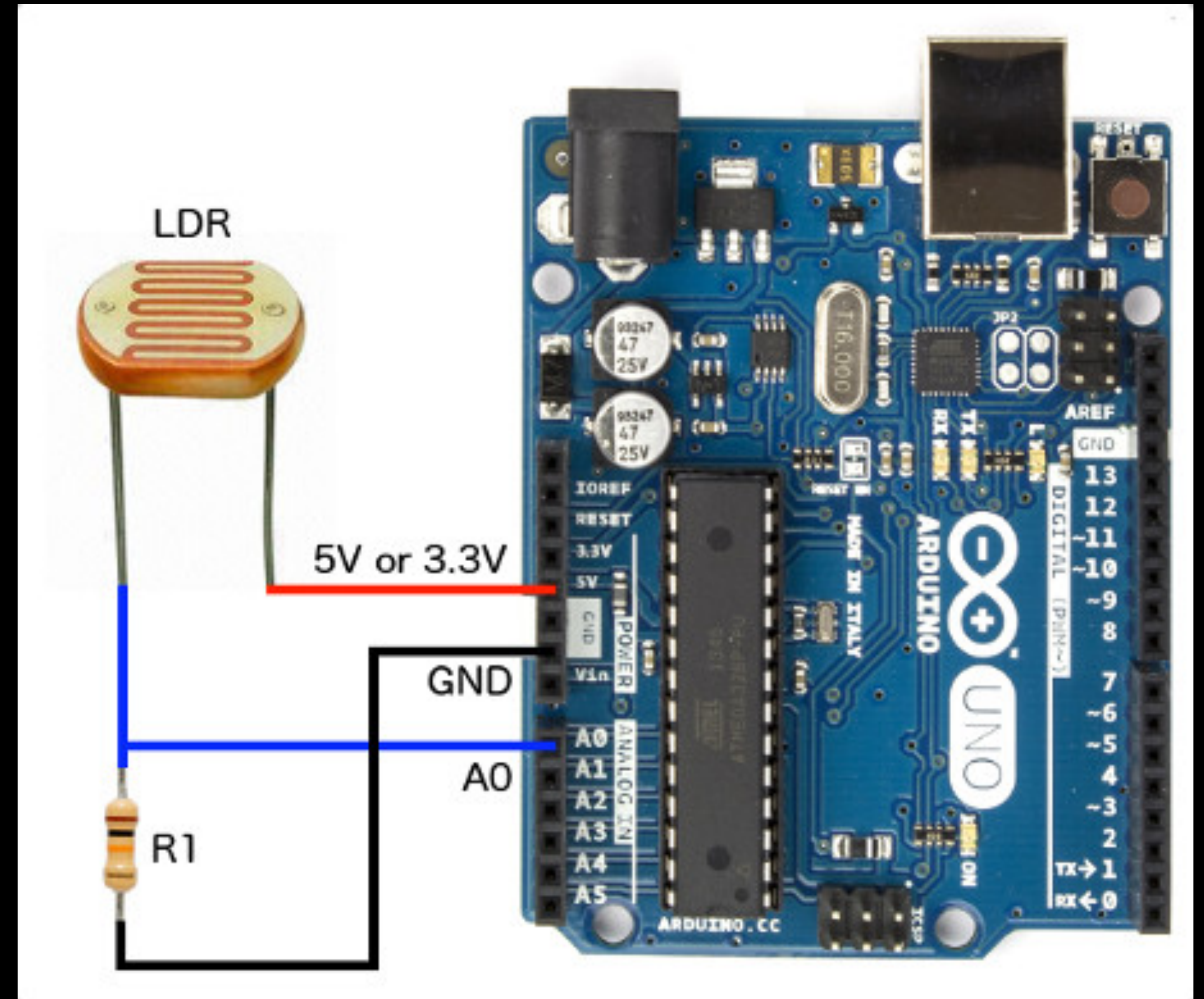
Setting up a circuit

- Pay attention to COLOR of wires in the instructions
- Read what is written on the board usually in white
- Take it one step at a time
- Double / triple check your work!

All analog sensors (Light, Flex, Pressure, knob)

- have 3 wires
- Red, Black, and (blue or yellow).
- Red > 5v
- Black > GND
- Yellow/Blue > A0

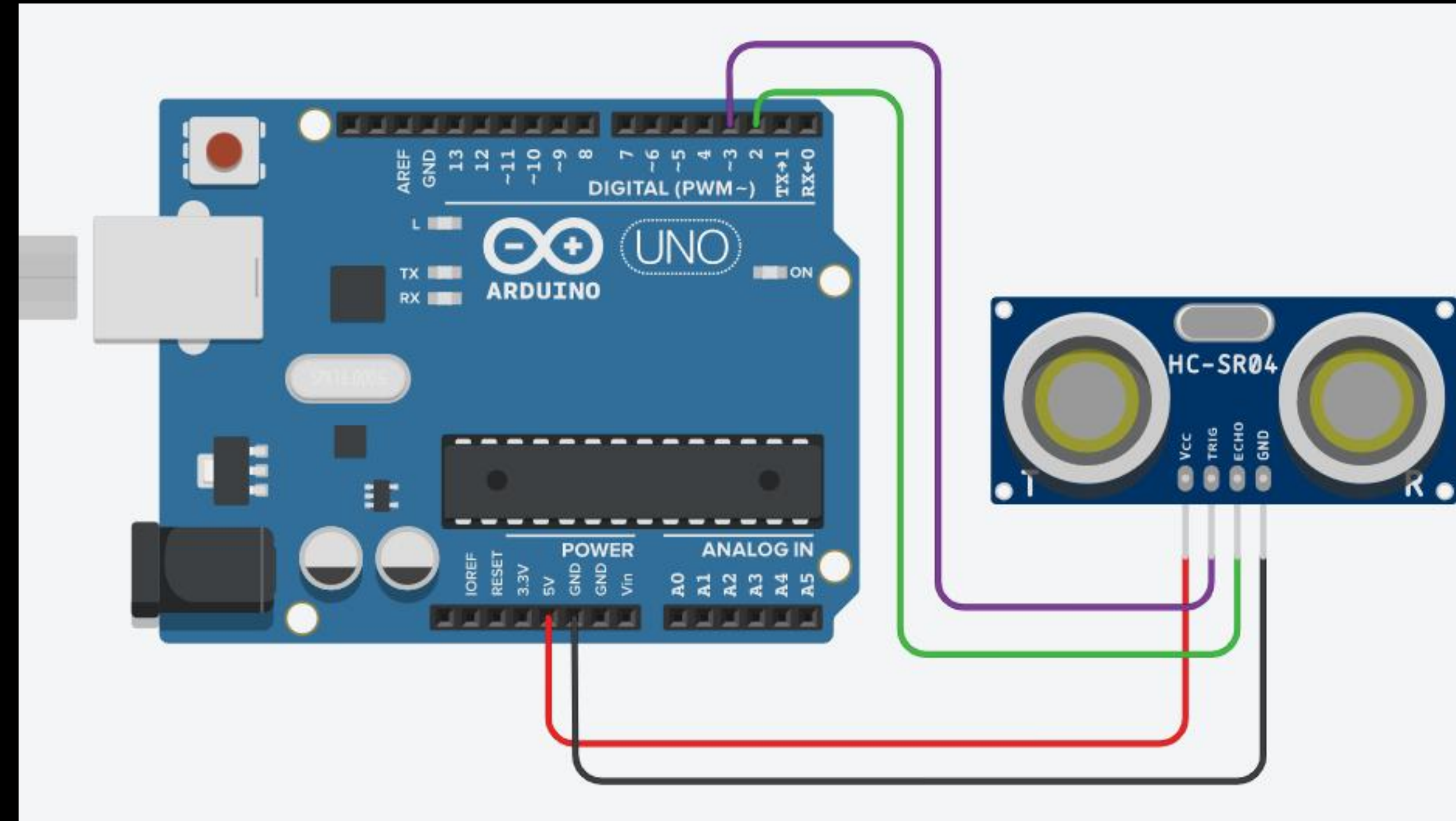
Output: You will get 1 number that indicates the value of the sensor



All analog sensors

- 4 wires
- Read the card writing
- GND > GND
- VCC > 5V
- TRIG > 3
- ECHO > 2

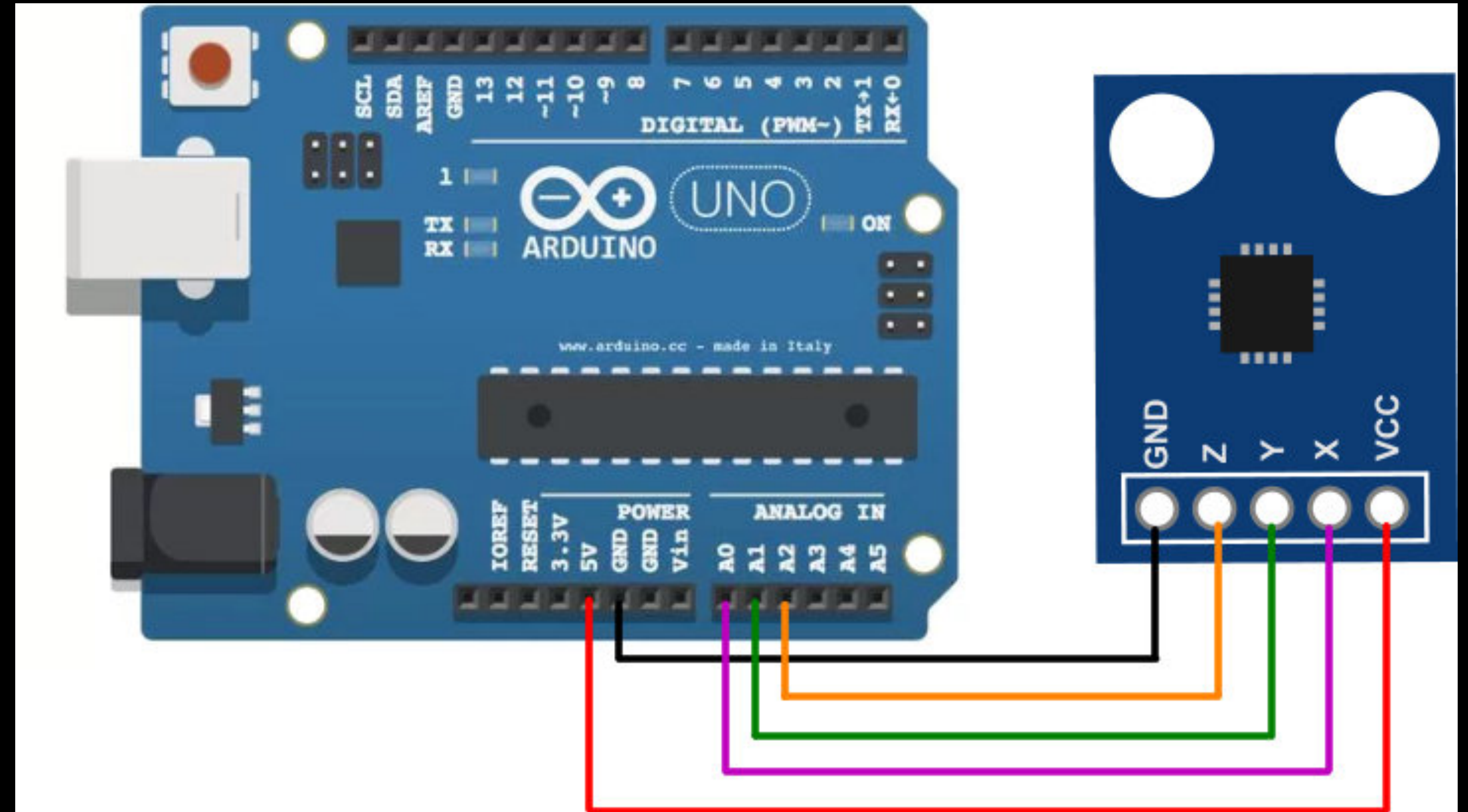
Output: you will get 1 number that indicates distance away from the sensor



All analog sensors

- 5 wires
- GND > GND
- VCC > 5V
- Z > A2
- Y > A1
- X > A0

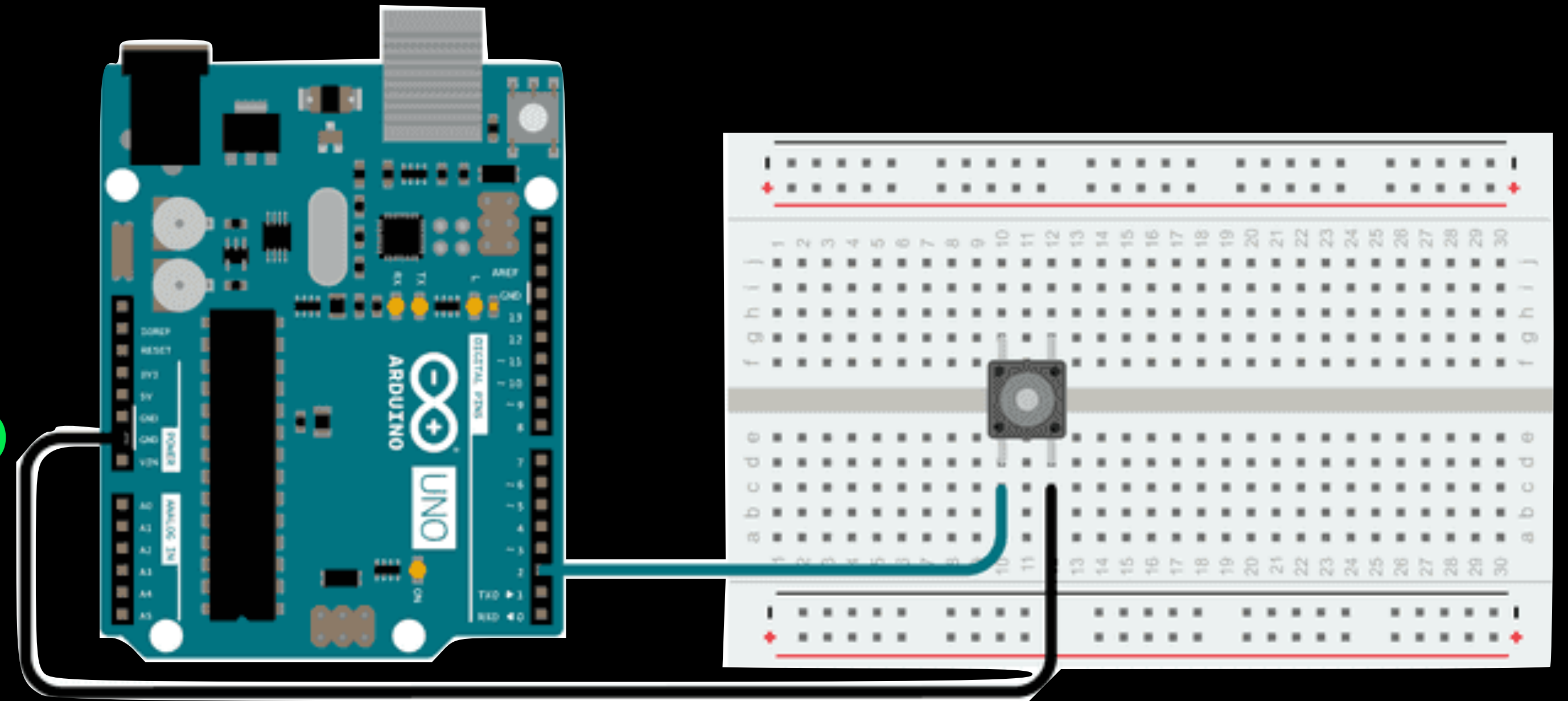
Output: You will get 3 numbers that correspond to the XYZ axis of rotation. Use the 3 number option in the maxpatch



Buttons

- GND
- 2

Buttons are just a broken connection, so feel free to use anything as a button!
More on this next week...



Output: You will get a 0 or 1 depending if the button is pressed or not

Code

- I've already uploaded the code onto each Arduino for you, but feel free to look at it and change it here:

[https://github.com/LeeCyborg/
Arduino-To-MaxMSP/tree/main](https://github.com/LeeCyborg/Arduino-To-MaxMSP/tree/main)

Download the Arduino IDE to load code at [arduino.cc](https://www.arduino.cc)

This is the pin your sensor is connected to

Open a serial port between the computer and Arduino at this rate

```
int analogPin = 0;
```

```
void setup(){  
  Serial.begin(9600);  
}
```

Print it to the Serial, this is what your Max Patch is reading!

```
void loop() {  
  Serial.println(analogRead(analogPin));  
  delay(100);  
}
```

Serial.print send to serial
analogRead read sensor value
analogPin the pin to read

Sensor Exploration

- Work in pairs or teams
- Explore sensors in different contexts
- Integrate into patches, or modify the Arduino code (if you're up for it!)
- Get at least 1 sensor working and integrate it into something creative

Some things to explore:

- How can you integrate sensors into an environment?
- How can sensors interact with the viewer ?
- Consider that you can extend wires to virtually any length, integrate them into existing or fabricated

Sensors for your final:

- You can buy your own, borrow from me, talk to James to use the equipment available to you. Arduino's are available to borrow.
 - More advanced and specific sensors might need to be purchased by you
 - No need to use Arduino in your final.
- In-class work can be done for your sensor exploration project.

Handing in projects

Each project should include a .zip file that contains:

You can also upload to GitHub in a repository and provide the link

- A saved file (File > Save as Project) .maxpatch
- A screen capture of your max patch
- A video or audio recording of your project working (can be a video or a link to a private video on YouTube, Vimeo or Google)
- A 50-100 word explanation of your project **inside your maxpatch**

Be sure to name files properly (no untitled-1.zip)

All files are expected to be cleaned up and arranged in a reasonable, legible way. Videos should be clear, well light and show your project working.

Homework:

- Start thinking about your final projects
(completely open project)

Next week:

More sensor exploration with DIY sensors